

Zarailink — Seller Query Framework

Below is a seller-side framework that mirrors your Buyer Query Framework but flipped to the seller's perspective. For each family I define: **what users typically ask**, **what the system extracts**, **system behaviour (how results are produced)**, and **what is shown on the Search Results page**. This assumes **seller intent** (user wants to find buyers / demand for a product).

Tip: keep the same numbering as Buyer families (1–9) so you can point-to-point compare for demos.

FAMILY 1: Simple Discovery (Base Case — Who buys X?)

What the user asks

- “Who buys dextrose?”
- “Buyers of dextrose”
- “Who purchases glucose syrup?”

System extracts

- Intent = SELL (find buyers)
- Product = dextrose / glucose syrup
- Optional filters: destination country (Pakistan, UAE), time window

System behavior

- Query transaction database for importers / buyer-side entities that have historically purchased this product (reverse of exporter lookup).
- Rank buyers by: total purchases of the product, purchase frequency, recency, average price paid.

- If buyer-side identity is ambiguous (e.g., intermediaries/brokers), surface both company and buyer-type tags.

What is shown

Primary

- Ranked buyer cards: Buyer name | Country | Typical purchase size | Avg price paid | Last purchase date | “Based on X purchases”
- Contact / lead button (if allowed)

Secondary

- Sample transaction table (recent deals)
 - Demand strength indicator (High / Medium / Low)
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FAMILY 2: Country-Filtered Buyer Discovery

What the user asks

- “Who buys dextrose in Pakistan?”
- “Buyers of sugar in UAE”
- “Potential buyers in India”

System extracts

- Product
- Destination country filter (one or many)
- Optional buyer type (trader, manufacturer, distributor)

System behavior

- Filter historical transactions by destination country.

- Rank buyers inside that country by product-specific purchase volume/frequency.
- Apply buyer-type filter to prioritize relevant buyer categories (e.g., wholesalers vs FMCG manufacturers).

What is shown

Primary

- Ranked buyer cards from that country only

Secondary

- Country snapshot: total demand (volume) for the product, number of buyers, avg buyer purchase size
 - Map / heatmap (if available) showing buyer concentration
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FAMILY 3: Volume-Aware Buyer Discovery (Match by Order Size)

What the user asks

- “Who buys 50MT of dextrose?”
- “Buyers looking for 1–5 ton shipments”

System extracts

- Product
- Target order volume (normalize to MT)
- Acceptable delivery windows

System behavior

- Compute volume-fit: compare requested size to historical purchase sizes per buyer.

- Boost buyers who frequently purchase similar sizes or who have repeatedly placed orders in that band.
- Optionally show buyers who consolidated multiple small purchases (for brokers/aggregators).

What is shown

Primary

- Ranked buyer cards with a **Volume-fit indicator** (e.g., Strong / Good / Weak match)
- Avg purchase size for buyer

Secondary

- Evidence table: sample past purchases of similar size
- Suggested packing/logistics notes for that volume bracket

FAMILY 4: Price-Constrained Buyer Discovery

What the user asks

- “Buyers who paid > \$700/MT”
- “Find buyers who paid premium prices for dextrose”
- “Buyers that accept 2% commission”

System extracts

- Product
- Price floor/ceiling or price expectation
- Payment / commercial terms (if specified)

System behavior

- Filter buyers by historical prices paid (or average accepted price).
- If none match exactly, return closest matches and explain gap.
- Rank primarily on price compatibility (buyers who historically accepted target price).

What is shown

Primary

- Ranked buyer cards with Avg price paid and Best recorded purchase price

Secondary

- Price distribution histogram for buyers
 - Advice card: “No exact matches — consider lowering price by X% or offering sample”
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FAMILY 5: Time-Constrained / Recency-Based Buyer Discovery

What the user asks

- “Who bought dextrose in the last 6 months?”
- “Which buyers were active this year?”

System extracts

- Product
- Time range filter

System behavior

- Filter buyer transactions to the specified time window.
- Rank by activity within window: frequency, volume, and recency.

- Penalize buyers inactive for long periods.

What is shown

Primary

- Ranked buyer cards with **Recency badge** (e.g., Active last 30 days)

Secondary

- Recent shipment table and buyer activity summary (e.g., “5 purchases in last 3 months”)
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FAMILY 6: Recommendation / Outreach (Who should I contact first?)

What the user asks

- “Which buyers should I contact first for 50MT?”
- “Top 3 buyers to approach”

System extracts

- Product
- Constraints (volume, price, country)
- Outreach goals (quick sale, premium price, long-term buyer)

System behavior

- Run a weighted recommendation aggregator combining volume-fit, price-fit, buyer reliability, and recency.
- Add outreach readiness signals (e.g., buyer recently purchased similar product, buyer frequency suggests repeat purchases).
- Produce shortlists tailored to outreach goals.

What is shown

Primary

- “Top N recommended buyers” module with a one-line rationale per buyer (e.g., “Bought 40–60MT monthly, paid premium prices”)
- Action buttons (Contact, Save Lead, Export CSV)

Secondary

- Full ranked buyer list below
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FAMILY 7: Market Opportunity / Country Comparison (Seller-Centric)

What the user asks

- “Which country has the highest demand for dextrose?”
- “Compare Pakistan vs Bangladesh buyers for dextrose”

System extracts

- Product
- Comparison countries (list)
- Metric preference (price, volume, buyer count)

System behavior

- Aggregate buyer-side demand metrics by destination country (total import volume, number of buyers, average price paid).
- Score countries by seller-relevant metrics (demand growth, price premium, ease-of-entry).

What is shown

Primary

- Country comparison table: total buyer-side volume, avg price paid, buyer count, demand growth %
Secondary
 - Top buyers per country, market-entry advice (logistics, tariffs, packaging norms)
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FAMILY 8: Transaction Evidence / Buyer Verification

What the user asks

- “Has Buyer X purchased dextrose before?”
- “Show shipments from Pakistan to Buyer Y”
- “Provide invoices / deal evidence for Buyer Z”

System extracts

- Product
- Buyer name (optional)
- Origin / destination / time range (optional)

System behavior

- Direct retrieval of historical buyer transactions; no ranking.
- Provide evidence: transaction rows, counterparty verification info (KYB-like badges if available), repeat-purchase indicators.
- If buyer has no record, optionally show similar buyers or brokers.

What is shown

Primary

- Full transaction table: Seller | Buyer | Qty | Price | Date | Shipping terms
Secondary

- Buyer summary card: total purchases, first purchase date, average price, verification status
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FAMILY 9: Hybrid / Multi-Intent Seller Queries

What the user asks

- “Find buyers in Pakistan who pay > \$700 and prefer 20–50MT”
- “Which countries buy dextrose most, and who are the top buyers there?”
- “Find buyers and suggest logistics partners”

System extracts

- Product
- Multiple constraints (volume, price, country)
- Multi-subintent detection (buyers + market comparison + logistics)

System behavior

- Decompose query into sub-intents; run each module (buyer discovery, country comparison, logistics suggestions).
- Stack modules in results with clear section labels and cross-links.
- If subintents conflict, show prioritized answers and explain tradeoffs.

What is shown

- Supplier-discovery block (buyers list)
- Country comparison block (if requested)
- Recommendation / Outreach block

- Logistics / trade notes block
 - Clear separation and explanation of which part answered which intent
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UNIVERSAL SELLER QUESTIONS (Across All Families)

Every SELL query ultimately answers some combination of:

1. **Who buys this product?** (list of buyers / leads)
2. **What price do buyers pay?** (avg, best recorded)
3. **What volumes do buyers purchase?** (typical order sizes)
4. **How active / reliable are these buyers?** (frequency, recency)
5. **Who should I contact first?** (shortlist + rationale)
6. **What market/country offers best opportunity?** (price vs demand)
7. **Is there verified transaction evidence?** (proof of past buys)

Design the UI and API to always surface these seven facets where applicable